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High Manganese and Aluminum Steels for the Military and Transportation Industry

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Lightweight advanced high strength steels (AHSS) with aluminum contents between 4 and 12 weight percent have been the subject of intense interest in the last decade because of an excellent combination of high strain rate toughness coupled with up to a 17% reduction in density. Fully austenitic cast steels with a nominal composition of Fe-30%Mn-9%Al-0.9%C are almost 15% less dense than quenched and tempered Cr-Mo steels (SAE 4130) with equivalent strengths and dynamic fracture toughness. This article serves as a review of the tensile and high-strain-rate fracture properties associated mainly with silicon additions to this base composition. In the solution-treated condition, cast steels have high work-hardening rates with elongations up to 64%, room-temperature Charpy V-notch (CVN) impact energies up to 200 J, and dynamic fracture toughness over 700 kJ/m^2. Silicon additions in the range of 0.59–1.56% Si have no significant effect on the mechanical properties of solution-treated steels but increased the tensile strength and hardness during aging. For steels aged at 530°C to an average hardness of 310 Brinell hardness number, HBW, increasing the amount of silicon from 1.07% to 1.56% decreased the room temperature CVN breaking energy from 92 J to 68 J and the dynamic fracture toughness from 376 kJ/m^2 to 265 kJ/m^2. Notch toughness is a strong function of phosphorus content, decreasing the solution-treated CVN impact toughness from 200 J in a 0.006% P steel to 28 J in a 0.07% P steel. For age-hardened steels with 1% Si, increasing levels of phosphorus from 0.001% to 0.043% decreased the dynamic fracture toughness from 376 kJ/m^2 to 100 kJ/m^2.

INTRODUCTION

High-manganese steels containing aluminum are derived from Robert Hadfield's original investigations of a Fe-13wt.%Mn-1.2wt.%C steel with high toughness and wear resistance.[1] All chemistry values are in weight percent unless otherwise specified. Increasing the manganese content to the range of 20–30% and adding aluminum in levels up to 10% produces a fully austenitic steel when solution treated above 950°C and reduces the density of these steels up to 15% when compared with conventional quench and tempered steels. Therefore, these lightweight steels may be of interest to the transportation industry as corporate average fuel economy (CAFE) is ratcheted up to 54 mpg by 2025.[2,3] The lightweight steels with compositions in the range of 20–30% Mn, 5–11% Al, and 0.3–1.2% C obtain their strength as a result of age hardening by the coherent precipitation of nanosized κ-carbide. These carbides precipitate homogeneously in the austenitic matrix.[2–6] Mechanical properties are a function of processing, composition, and age hardening. Figure 1 compares representative microstructures of as-cast and hot-rolled and recrystallized fully austenitic lightweight steels. As-cast alloys consist of large primary austenitic dendrites that produce grain structures on the order of millimeters. Hot rolling refines the grain size, and the steel shown in Fig. 1b consists of fine equiaxed austenite grains and annealing twins. Hot- and cold-rolled solution-treated steels have been shown to have exceptional strengths and toughness with strengths as high as 2 GPa and greater than 80% true fracture strain[7,8] In age-hardened steels, strengths as high as 1210 MPa have

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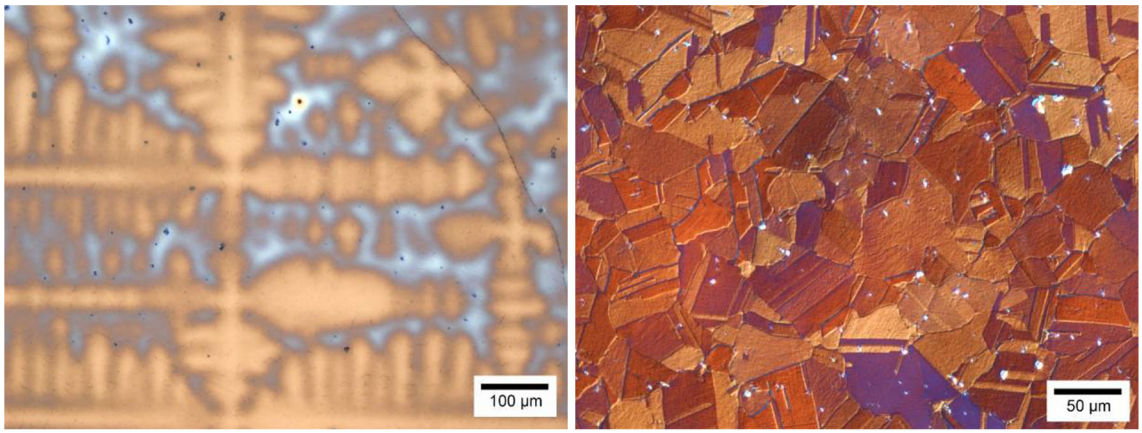


Fig. 1. (a) Bright-field optical micrograph of an as-cast Fe-28Mn-3Al-0.9Si-1.2C-0.6Mo steel with a microstructure of large primary austenite dendrites. (b) In contrast, the hot-rolled Fe-29Mn-6Al-0.9C-0.29Si-0.44Nb-0.44Mo steel consists of equiaxed austenite grains and annealing twins that are much smaller with an average grain size of 50 μm and are pinned by NbC on grain boundaries. The specimen in (b) was imaged using differential interference contrast imaging. Both specimens were etched using 10% nital.

been reported by Acselrad et al.[9] for a wrought Fe-29Mn-8.8Al-1C-1.33Si steel. Cast lightweight steels have low melting points, less than 1350°C, and excellent filling characteristics that are similar to cast irons.[2] Recent studies of as-cast and age-hardened Fe-30Mn-9Al-0.9C-(1–1.4)Si steels have shown that these steels are competitive with quenched and tempered Cr-Mo steels and can attain as high as 1085 MPa in tensile strength with greater than 400 kJ/m^2 of dynamic fracture toughness, depending on the degree of age hardening.[10,11]

Recent research has focused on slight alloy adjustments, mainly silicon, to nominal composition Fe-30Mn-9Al-0.9C steels for use in the transportation industry, for military armor plate, as well as for wear-resistant applications.[3,10–16] However, the high manganese content (20–30%) required to stabilize a high aluminum austenitic matrix may relegate these grades of steel to castings or casting preforms that can be forged. Current steelmaking practices for wrought product are limited to manganese contents less than 8%, although Pohang Iron and Steel Company (POSCO) has been successful in producing second-generation advanced high-strength steels that exhibit twin-induced plasticity (TWIP) with 15–18% Mn. However, electrolytic manganese is often required for these higher manganese steels to limit temperature losses after manganese addition, which drives the cost to that of the Cr-Ni stainless steel grades.

Silicon is added to austenitic Fe-30Mn-9Al-0.9C steel for a variety of different reasons. Silicon additions increase the fluidity and decrease the melting point by 30°C/wt.% Si, allowing for greater flexibility of design and good filling of complex shapes and thin sections.[2] Additionally, silicon has been reported to prevent or suppress the formation of brittle β-Mn during extended age hardening, which is detrimental to ductility and toughness.[17]

Table I. Steel chemistries in weight percent

Si	C	Mn	P	S	Mo	Al	Cu
1.01	0.89	30.20	0.006	0.001	0.31	8.85	–
0.92	0.94	29.10	0.043	0.006	0.33	8.28	0.09
1.07	0.90	29.11	0.070	–	0.37	9.25	0.10
0.59	0.95	30.35	0.002	0.006	0.54	8.74	0.01
1.07	0.90	30.42	0.001	0.006	0.53	8.83	0.006
1.56	0.89	29.97	0.002	0.007	0.53	8.81	0.006

Adding silicon has also been shown to increase the strength and hardness during aging but decrease work-hardening rates.[10,11,18]

High-strain-rate fracture in these lightweight steels is also a function of phosphorus content and melt cleanliness. Phosphorus in levels greater than 0.06% P greatly increases the kinetics of age hardening but sharply decreases the Charpy V-notch (CVN) impact toughness by as much as 80%.[13,16] It is well known that melting practice contributes to fracture because it affects the composition, size, distribution, and morphology of inclusions in steel. Schulte et al.[19] have shown that the loss in CVN toughness in a Fe-30Mn-9Al-1Si-0.9C steel is strongly correlated with increasing amounts of AlN inclusions, which reduced the CVN breaking energy at −40°C by almost 50%.

This article serves a review of the tensile and high-strain-rate fracture properties associated with cast and wrought steels with a nominal composition of Fe-30Mn-9Al-0.9C. The effect of silicon and phosphorus on strength and toughness during aging is reviewed with respect to new and previously unpublished findings. Table I gives the chemistry (wt.%) of the respective steels discussed in this article.

AGE HARDENING

Increases in the strength and hardness of these lightweight steels are provided by the homogeneous

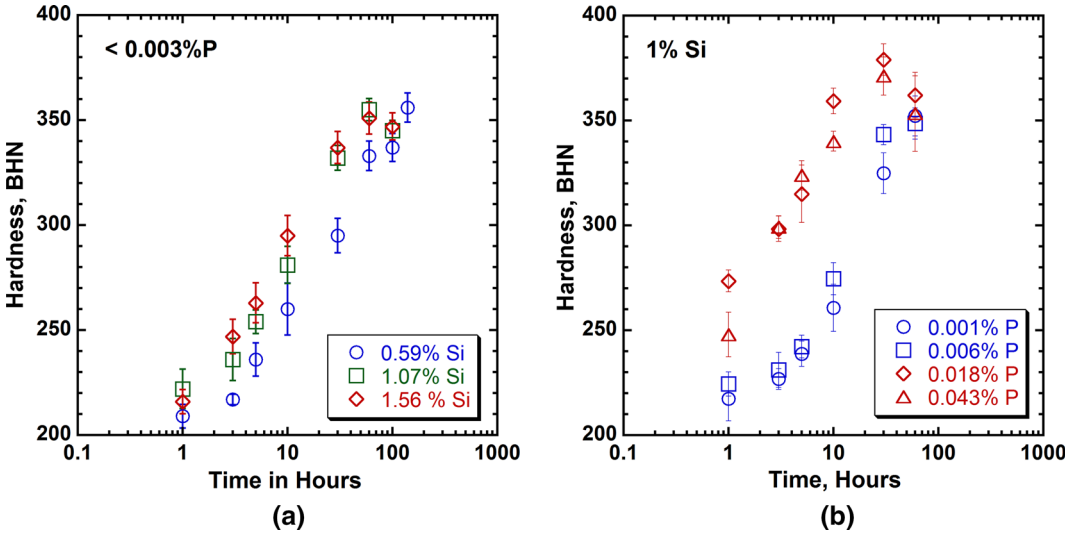


Fig. 2. The 530°C age-hardening curves for (a) low-phosphorus Fe-30Mn-9Al-XSi-0.9C steels and (b) Fe-30Mn-9Al-1Si-0.9C steels with between 0.001% and 0.043% P show that both Si and P increase hardness during aging.[16]

and coherent precipitation of nanosized κ-carbide, $(Fe,Mn)_3AlC_x$, which has the $E2_1$ cubic perovskite crystal structure and is deficient in carbon, i.e., $x < 1$. κ-Carbide has a cube-on-cube orientation relationship with the austenite matrix: $\langle 100\rangle_\kappa // \langle 100\rangle_\gamma$ and $\{001\}_\kappa // \{001\}_\gamma$.[6,7,15,17,20,21] Most studies indicate that precipitation of κ-carbide in the temperature range of 450–700°C is the result of compositional modulation produced by spinodal decomposition into either carbon rich[9,21–23] or carbon and aluminum rich[6,7,17,24] and depleted zones. Spinodal decomposition is thought to either precede,[7] take place concurrently to,[6,9,21,23–25] or subsequently to[17,22] short-range ordering (SRO) into lattice sites corresponding to the κ-carbide structure. The initial stage of hardening occurs very rapidly and spinodal decomposition and ordering has been reported to take place during quenching from the solution treatment temperature.[24,26,27] Compositional modulation and ordering is followed by coarsening with the development of cuboidal shaped precipitates and by transitioning to plate morphologies, periodically arranged along elastically soft $\langle 100\rangle$. Silicon and phosphorus have been shown to increase the hardening effect during aging as shown in Fig. 2 for Fe-30Mn-9Al-XSi-0.9C steels aged at 530°C.[16] Figure 2a shows the hardening effect of silicon in a low-phosphorus, <0.003% P, steel. Increasing the amount of silicon from 0.59% to 1.56% had no effect on the solution treated hardness but increased the hardness during aging without significantly increasing the peak hardness. On the other hand, phosphorus is shown to greatly increase both the hardening rate and the peak hardness during aging of a 1% Si steel as shown in Fig. 2b.

The mechanism behind the hardening effect with silicon addition has only recently been resolved by Bartlett et al. for Fe-30Mn-9Al-(0.59–1.56)Si-0.9C-0.5Mo cast steels.[18] This study used a combination of transmission electron microscopy (TEM), 3-D atom probe tomography (APT), x-ray diffraction (XRD), and first principles modeling to study the precipitation. In solution-treated specimens, XRD and TEM results confirmed that, regardless of Si content, no spinodal decomposition or short-range ordering within the austenite had taken place. However, increasing the amount of silicon accelerated formation of κ-carbide precipitates as shown in Fig. 3 and appears to increase the volume fraction. Silicon was shown by ab initio calculations and experimental results to increase the activity of carbon in austenite and stabilize the κ-carbide at higher temperatures.[18] The repulsive Si-C interaction was responsible for increasing the partitioning coefficient of carbon in κ-carbide from 2.1 to 2.9 in 0.59% and 1.56% Si steels aged 60 h at 530°C.[18] Silicon was shown to partition to austenite during aging and decrease the austenite lattice parameter. Thus, as the silicon content of the austenite increases, the repulsive interaction between silicon and carbon forces an increase in carbon content of the κ-carbide to produce a corresponding increase in the κ-carbide lattice parameter. The respective partitioning of carbon and silicon increases the γ–κ constrained mismatch and increases the hardening effect. The predicted increase in the yield strength,[18] as determined using a spinodal hardening mechanism derived from the work of Kato et al.[28] was calculated to be 120 MPa/wt.% Si for specimens aged at 530°C for 60 h, and this is in reasonable agreement with previously reported mechanical properties.[11]

Phosphorus in levels greater than 0.006% P is shown to have a much more pronounced effect on age hardening than silicon addition, increasing both the hardening rate as well as the maximum aged hardness as shown in Fig. 2b. However, phosphorus in amounts greater than 0.006% has been shown to

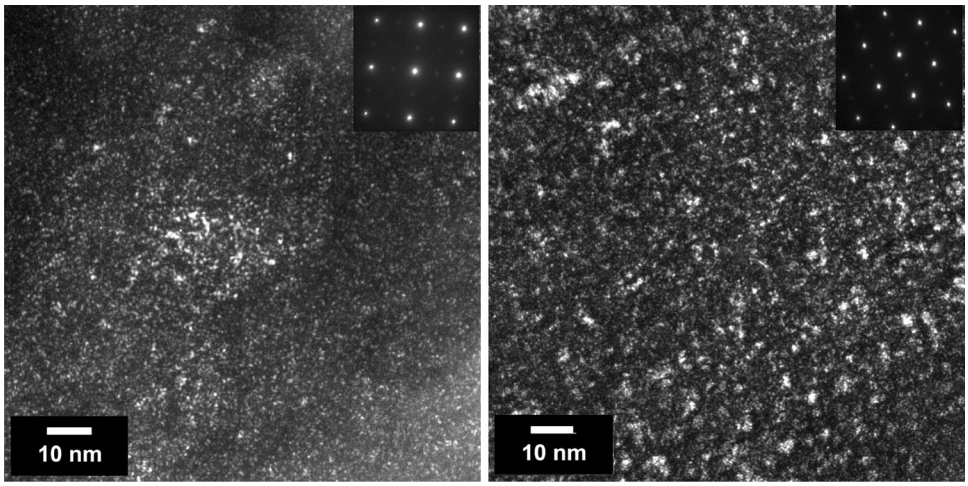


Fig. 3. Dark-field images of the (a) 0.59% Si and (b) 1.56% Si steels aged for 63 h at 490°C show that the size and number density of κ-carbides appear greater with increasing silicon.

decrease notch toughness by up to 80% in an aged Fe-30Mn-9Al-1Si-0.9C-0.5Mo steel.[11,13]

First-principles calculations have shown that substitution of phosphorus for aluminum positions in the κ-carbide is most favorable, and this is predicted to create open-volume defects on {100} when a stress is applied normal to the plane. Phosphorus substitution for aluminum in the κ-carbide structure as previously reported by Medvedeva et al.[29] relaxes the local bonding character along the $\langle 100 \rangle \kappa$. This results in a 45% reduction in the cleavage stress of the κ-carbide and is predicted to be the mechanism behind the reduction in notch toughness of aged Fe-Mn-Al-C steels containing phosphorus.[29] In more recent work,[30] the effect of phosphorus on κ-carbide precipitation in Fe-30Mn-9Al-1Si-0.9C-0.5Mo cast steels was investigated using a combination of TEM, three-dimensional APT, and XRD. Phosphorus was shown to increase the prevalence of the ordering reaction and high phosphorus, 0.043% P, specimens exhibited SRO that took place during the quench and prior to spinodal decomposition of the austenite. First-principles calculations show that SRO of Fe-Al-C into the $E2_1$ crystal structure is energetically favorable over a homogeneous solid solution.[31] In contrast, neither SRO nor spinodal decomposition took place during the quench in the previous study using low-phosphorus specimens.[18]

Phosphorus greatly increased the size and volume fraction of κ-carbide precipitation and accelerated both the first-stage ordering reaction as well as concurrent spinodal decomposition. Figure 4 shows the dark-field TEM $\langle 100 \rangle_\kappa$ images of a 0.001% P and 0.043% P Fe-30Mn-9Al-1Si-0.9C-0.5Mo steel aged for 0.5 h at 570°C. In the low-phosphorus steel, there is a high number density of small (a few unit cells in dimension) ordered domains that exhibit diffraction intensity corresponding to the perovskite crystal structure of κ-carbide (see Fig. 4a). In the high-phosphorus steel, these ordered domains are much larger at an average size of 5 nm and are preferentially coarsening along $\langle 100 \rangle \gamma$ as shown in Fig. 4b. One of the most important results from this study is that phosphorus was found to be randomly distributed between the matrix and the κ-carbide, and the increase in the rate of hardening with increasing phosphorus content is therefore not related to long-range diffusion of phosphorus. Instead, it was found that phosphorus greatly increases the rate of spinodal decomposition during aging at 530°C as shown in Fig. 5.

Increasing levels of P from 0.006% to 0.043% P increased the diffusion of carbon during aging, and after 10 h at 530°C, the carbon amplitude varied from 4.4 at.% to 6.0 at.%, respectively. It is therefore suggested that phosphorus increases the initial ordering reaction and that this greatly accelerates the later stage of spinodal decomposition. First-principles calculations show that phosphorus substitutions for aluminum corner positions in κ-carbide are most energetically favorable, and this may be responsible for the enhancement of the ordering reaction in the high-phosphorus alloys.[29]

A random walk estimate of the SRO kinetics of phosphorus atoms and aluminum atoms into κ-carbide corner positions during quenching was calculated. It was found that SRO of phosphorus into $E2_1$ corner positions of the κ-carbide structure may be an order of magnitude faster than that of aluminum. It should be noted that thermodynamic preference of phosphorus substitution for aluminum corner positions was not taken into account in the above calculation.[30] It is shown from the above review that increasing both silicon and impurity levels of phosphorus increases the kinetics of κ-carbide precipitation, the amount of carbon partitioning, and thus the degree of spinodal hardening during aging. The remainder of this article focuses on the tensile and high-strain-rate fracture properties associated mainly with different levels of Si and P.

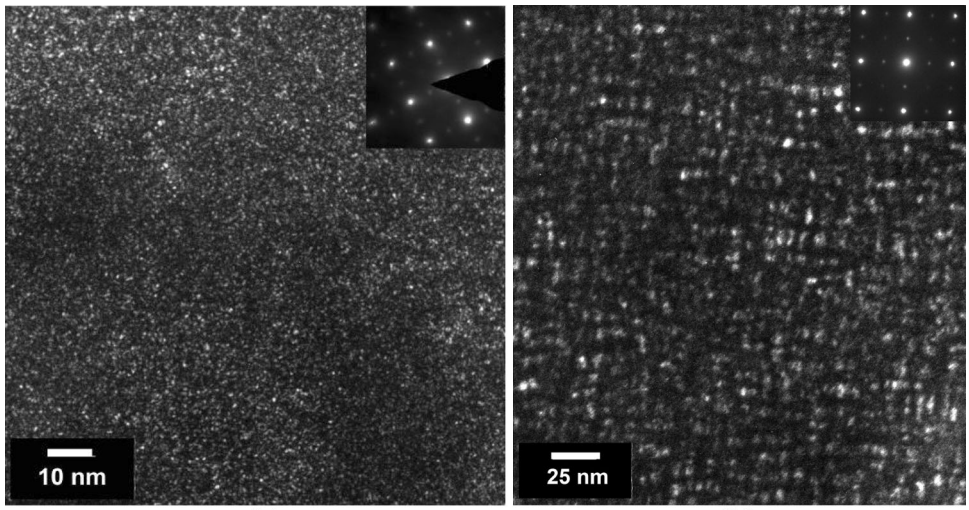


Fig. 4. Dark-field images of the (a) 0.001% P and (b) 0.043% P steels aged for 0.5 h at 570°C show that the size and number density of κ-carbides appear greater with increasing phosphorus.

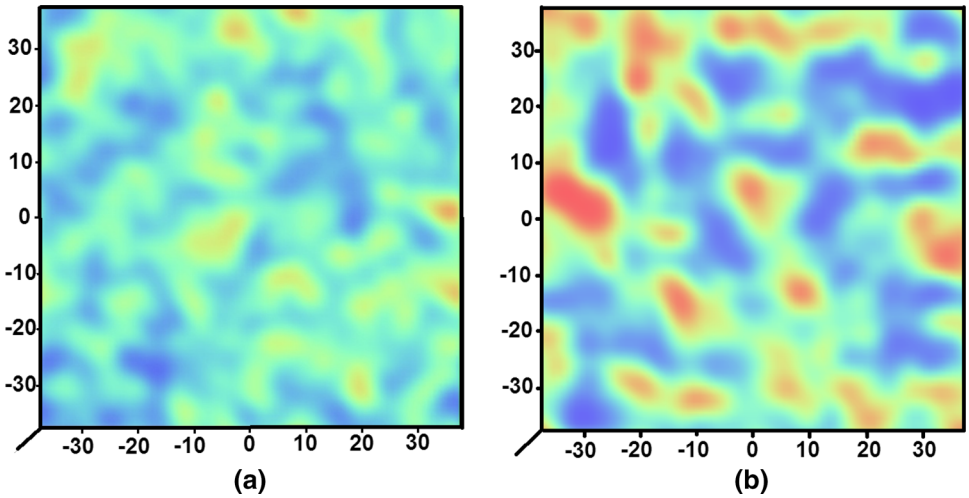


Fig. 5. The carbon concentration (2–6 at.% C—blue to red) maps obtained from APT analysis show the progress of spinodal decomposition in (a) 0.006%P and (b) 0.043%P steels aged for 10 h at 530°C. The scale is shown in nanometers (Color figure online).[30]

TENSILE PROPERTIES

Tensile properties have been shown to depend on the amount of silicon, the amount of phosphorus, and the cleanliness of the castings. Tensile properties are reported in Table II for steels of nominal composition Fe-30Mn-9Al-XSi-0.9C-0.5Mo and are comparable with the tensile properties of quenched and tempered SAE 4130 cast steel. Calcium treatment and improved foundry practices to eliminate dross have substantially decreased the nonmetallic inclusion density[19,32,33] and increased the ductility as shown in Table II. The increases in strength provided by the precipitation of κ-carbide are accompanied by a corresponding loss in work-hardening ability and a reduction in tensile ductility.[14] Dastur and Leslie[34] proposed that the high work-hardening rates associated with Hadfield steels is the result of dynamic strain aging associated with the formation of Mn-C dipoles in the solid-solution austenite and recent first-principles calculations by Medvedeva et al.[35] have confirmed that Mn-C dipoles are energetically favorable. However, the addition of aluminum may result in κ-carbide precipitation, which decreases the amount of carbon in solution, contributes to planar slip, and lowers work hardening rates.[14]

Increasing the silicon content from 1.07 to 1.56% had no effect on the as-solution treated mechanical properties but increased yield and tensile strengths from 873 to 937 MPa and from 953 to 1016 MPa in low phosphorus steels (< 0.007% P) aged 60 hrs at 530°C (see Table II).[19] However, increasing the silicon content from 1.07% to 1.56% decreased the elongation to failure from 20% to 13% in age-hardened steels. Increasing phosphorus levels from 0.001% to 0.06% P in 1% Si steels increased the 0.2% offset yield

Table II. Tensile properties cast Fe-30Mn-9Al-XSi-0.9C steels with different melt practices[11,12]

Wt.% Si	Wt.% P	Heat treatment	Hardness Brinell[a] No Calcium treatment	0.2% YS (MPa)	UTS (MPa)	% elongation[b]
		Solution treated at 1050°C				
1.0	0.06	2 h	224	458	687	44
1.4	0.06	2 h	219	549	766	36
		Aged at 530°C				
1.0	0.06	10 h	343	891	940	18
1.4	0.06	10 h	353	922	995	14
1.0	0.06	30 h	372	1005	1065	8
1.4	0.06	30 h	384	1016	1085	5
		Calcium treated with Ar stirring				
0.78	0.007	**As-cast**		616	698	17
		Solution treated at 1050°C				
1.07	0.001	2 h	197	452	737	64
1.56	0.002	2 h	208	450	733	64
		Aged at 530°C				
1.07	0.001	26 h	303 ± 7	728 ± 13	795 ± 11	27.7 ± 6.6
1.56	0.002	20 h	309 ± 5	800 ± 21	834 ± 20	30.2 ± 9.6
1.07	0.001	60 h	350 ± 3	873 ± 17	953 ± 25	20.1 ± 10.1
1.56	0.001	60 h	360 ± 6	937 ± 8.9	1016 ± 2.5	13.2 ± 4.2
		Calcium treated only				
		Aged at 530°C				
1.01	0.006	22 h	307 ± 10	678 ± 12	769 ± 18	35.8 ± 5.3
1.01	0.006	60 h	352 ± 4	907	999	18
		Aged at 570°C				
		4 h	311 ± 7	666 ± 37	757 ± 27	30.4 ± 16
1.01	0.006	10 h	340 ± 2	804	877	15.7

UTS ultimate tensile strength; *YS* yield strength. Uncertainty reported for test sample greater than two and represents one standard deviation. [a]Hardness Brinell (3000 kg) were converted from Rockwell B and C scale measurements. [b]Percent elongation was measured in a 25.4-mm (1-in.) gage length.

and ultimate strengths of the age-hardened steels but only increased the yield strength of the solution-treated material. Increasing phosphorus content decreased the total elongation in the solution-treated and aged steels. Therefore, increases in strength during aging with increasing silicon and impurity levels of phosphorus are directly related to an increase in the amount and carbon content of κ-carbide as shown by APT. The observation that phosphorus increases the yield strength and decreases ductility even in the solution-treated condition is likely a result of SRO during the quench.

The wrought tensile properties of a calcium-treated, low-phosphorus (0.006% P) Fe-30Mn-9Al-1.01Si-0.9C-0.5Mo steel were evaluated by Bartlett et al.[11] Cast plates were hot rolled at 900°C to an 84% reduction in thickness, solution treated for 10 min at 900°C, and then water quenched to produce an equiaxed grain structure that was 10 μm to 20 μm in average diameter as shown in Fig. 6. A comparison of cast and wrought tensile strength as a function of aging time are reported in Fig. 7. At equivalent ultimate strengths, grain refinement from hot rolling produced better ductility than the cast material (Fig. 7). Tensile strength in the wrought alloy increased by 200 MPa with aging, but peak aging produced κ-carbide precipitation on grain boundaries and lowered elongation from 54% to 14% (see Fig. 6).

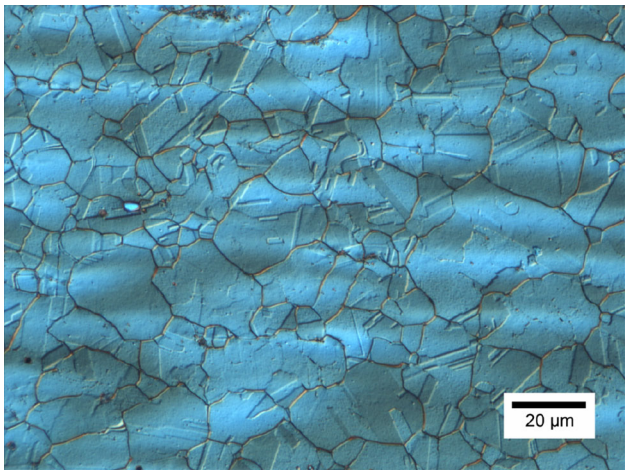


Fig. 6. The microstructure of the hot-rolled Fe-30Mn-9Al-1.01Si-0.9C-0.5Mo alloy in the peak aged condition (60 h at 530°C) shows preferential grain boundary etching due to κ-carbide precipitation as well as banding resulting from alloy segregation.[11] Specimen imaged using differential interference contrast techniques and etched with nital.

CHARPY V NOTCH TOUGHNESS

Notch toughness of Fe-Mn-Al cast steel is a strong function of phosphorus content. Table III gives the solution-treated notch toughness as a function of phosphorus content in steels with a nominal composition of Fe-30Mn-9Al-1Si-0.9C. Solution-treated notch toughness was found to decrease from 200 J to 28 J as phosphorus increased from 0.001% to 0.07%. As the amount of phosphorus increased in the solution-treated specimens the fracture mode changed from ductile failure in the 0.001% P samples to a low-energy cleavage-like fracture producing a dendritic pattern as a result of fracture along {001}in samples with 0.07% P as shown in Fig. 8. Again, the SRO of phosphorus is likely responsible for lowering the cleavage strength of {001} as predicted by Medvedeva et al.[29]

Slow quenching after solution treatment will also decrease notch toughness as a result of heterogeneous precipitation of κ-carbide along austenite grain boundaries.[33] The quench sensitivity of these

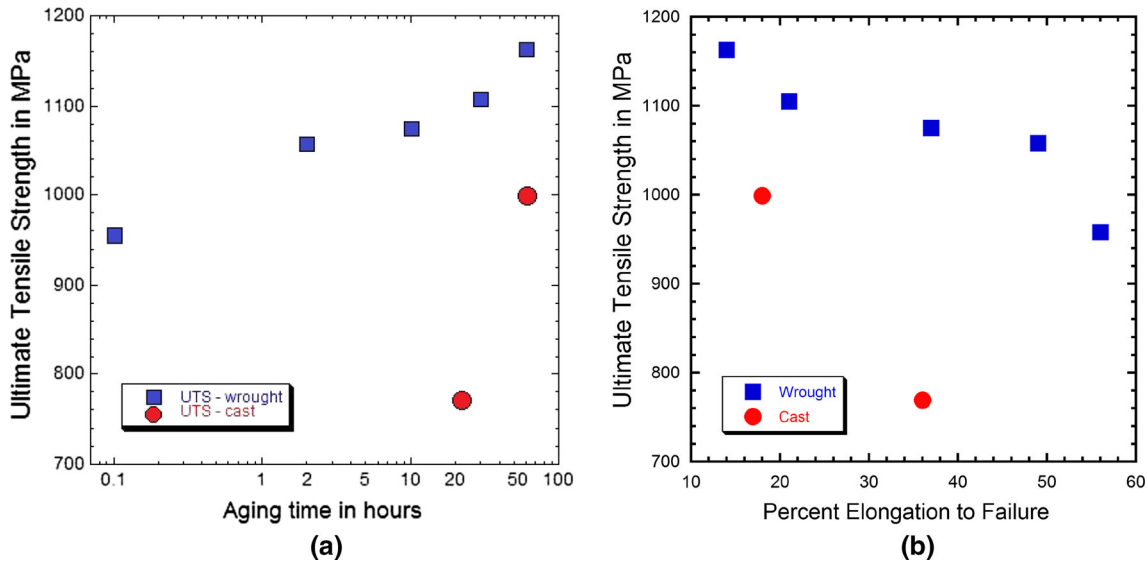


Fig. 7. (a) At equivalent aging times, refinement of the grain structure in the wrought alloy produced an increase in tensile strength when compared with the cast alloy. Results shown at 0.1 h are for the solution-treated material.[11]

Table III. The solution-treated CVN breaking energy for nominally Fe-30Mn-9Al-1Si-0.9C steels with different amounts of phosphorus

% Si	% P	Heat treatment	Hardness (BHN)	Average CVN energy (ft-lbs)	Average CVN energy (J)	Test temperature (°C)
1.07	0.07	Solution treated	242 ± 6	21 ± 4	28	25
1.01	0.006	Solution treated	194 ± 2	92 ± 1	124	25
1.07	0.001	Solution treated	192 ± 10	147 ± 17	199	25

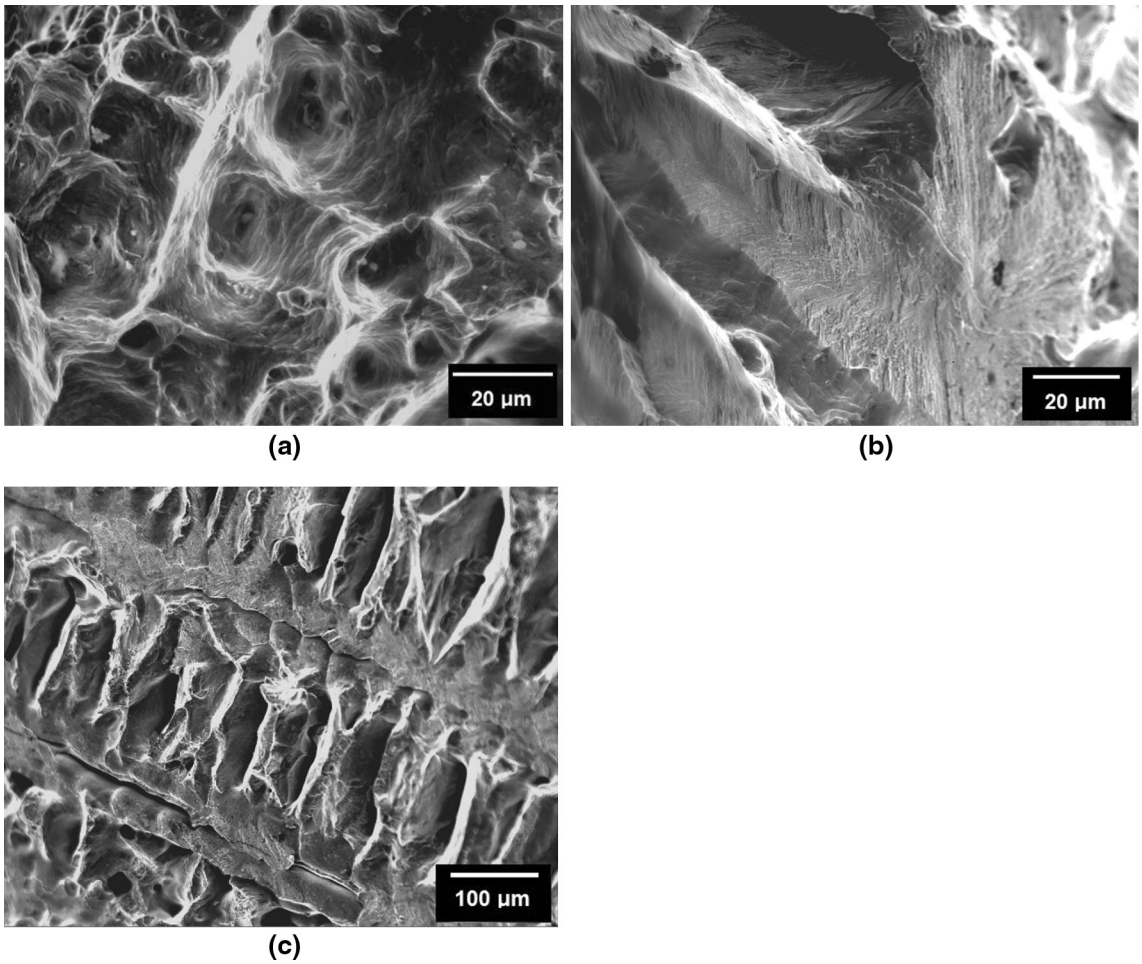


Fig. 8. The fracture surfaces of the as-solution treated and quenched (a) 0.001%P and (b) 0.07% P Charpy V-notch specimens show a transition in the fracture mode from ductile tearing in (a) to a low-energy cleavage-like fracture producing a dendritic pattern in (b and c).

lightweight steels will also be increased by segregation of phosphorus during solidification and the phosphorus will promote precipitation of κ-carbide along the austenite grain boundaries. The effect of phosphorus on the notch toughness of Fe-30Mn-9Al-1Si-0.9C steels aged for 10 h at 530°C is shown in Fig. 9.[11,13] Phosphorus increases the hardness of aged steels but dramatically decreases the notch toughness by more than 80%. Schulte et al.[19] have shown that it is possible to mitigate the effects of phosphorus by adding misch metal (predominately Ce and La) during furnace melting.[19] Inclusion analysis was performed using an ASPEX scanning electron microscope (SEM; FEI Company, Hillsboro, OR) with PICA 1020 automated feature analysis software to determine how inclusion type and distribution affects the CVN impact energy of Fe-Mn-Al-C alloys.

Schulte et al.[19] show that Ce and La combine with P to form complex inclusions, indicating that the phosphorus embrittlement of Fe-Mn-Al-C steels may be mitigated by misch metal treatment. The room-temperature CVN impact energies of an aged Fe-30Mn-9Al-1Si-0.9C-0.3Mo alloy increased from

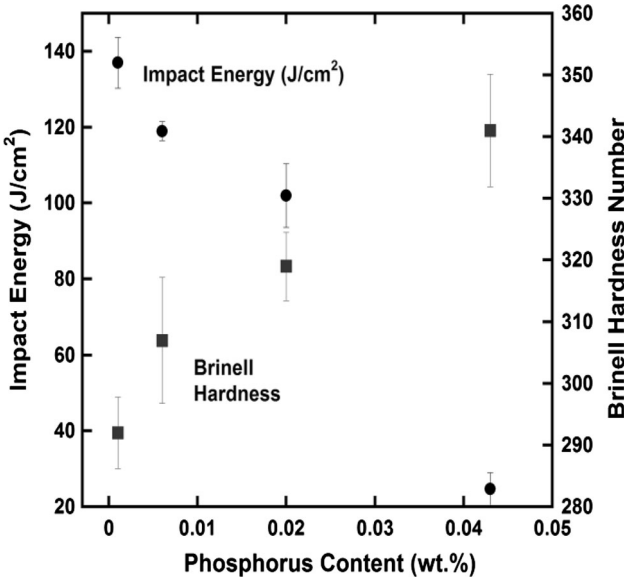


Fig. 9. Charpy V-notch impact energy at room temperature and hardness versus phosphorus content in the age-hardened (10 h at 530°C) condition. The figure shows a decrease in impact energy and increased aged hardness with increasing phosphorus content.[11,13]

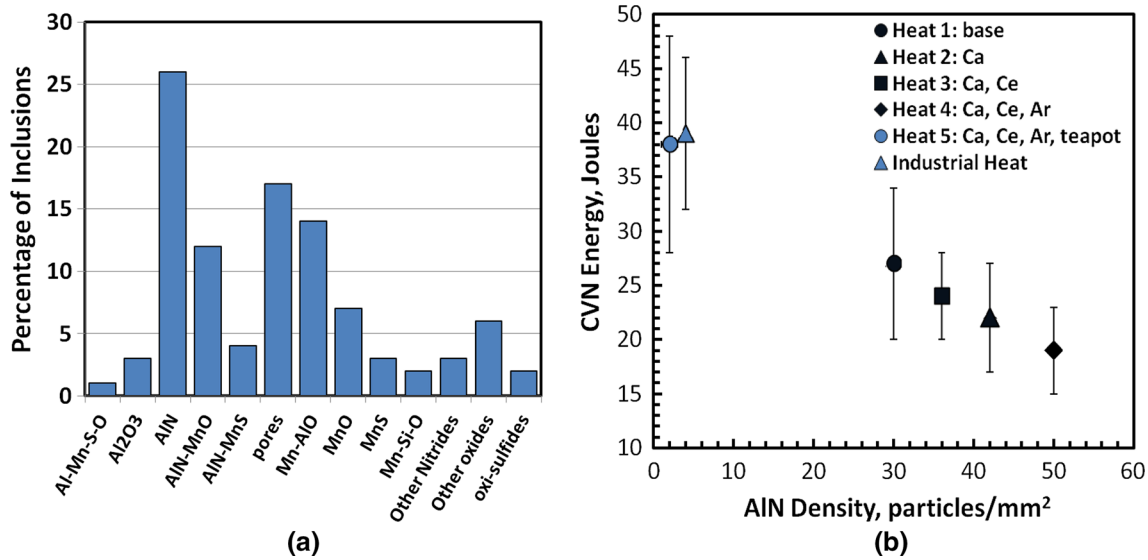


Fig. 10. (a) Inclusion analysis of a 0.001% P alloy that was calcium treated and argon stirred shows a high percentage of AlN. (b) A high AlN density was shown to be deleterious to impact properties. AlN content can be reduced with argon stirring.[19]

33 J, in the untreated condition, to 108 J with calcium and misch metal additions followed by argon stirring. In these calcium and misch metal treated steels, AlN inclusion density was the microstructural parameter showing the greatest correlation to the −40°C CVN impact energy. All other factors, including particle size, total inclusion density, and P content, were interrelated.[19] Argon stirring after Ca and misch metal treatment has the added benefit of removing AlN inclusions and improving notch toughness as shown in Fig. 10.

Unlike most austenitic steels, high-manganese steels have been shown by several researchers to have a ductile-to-brittle transition temperature with cleavage-like fracture at low temperatures.[13,36–39] It has been shown that the mechanism behind brittle fracture of the austenite at cryogenic temperatures is a combination of the strong temperature dependence of the yield strength and either brittle crack nucleation at the intersection of deformation twins or by accumulation of planar slip-induced dislocation defects on {111} and tensile separation of glide planes producing distinctive slip traces on cleavage facets.[36–39] The effect of silicon additions and testing temperature on notch toughness was evaluated for a Fe-30Mn-9Al-(1.07–1.56Si)-0.9C steel with low phosphorus (<0.003% P) aged for different times at 530°C. Figure 11 shows the ductile-to-brittle transition curves for the 1.07% and 1.56% Si specimens aged in the range of 300–352 BHN. Increasing the amount of silicon decreased the room temperature breaking energy from 93 J to 68 J in 12-h aged samples and from 58 J to 35 J in 30-h aged samples. Lowering the testing temperature to −40°C sharply reduced notch toughness and produced mainly brittle fracture with increasing silicon content lowering the breaking energy by an average of 15 J for samples aged for 12 h. The fracture surfaces of the 12-h aged specimens that were broken at −40°C are shown in Fig. 12 and reveal mainly brittle transgranular fracture with some favorably oriented cleavage facets exhibiting sets of slip traces in both silicon-containing specimens. Acselrad et al.[40] reported that aging produced a brittle transgranular cleavage-like fracture in which cleavage facets were found to contain linear markings along two or three different directions. They proposed that the mechanism of brittle cleavage fracture with aging was the result of "slipping off," or glide plane decohesion, along the slip plane producing a {111}-type fracture facilitated by planar slip and shear strain localization in shear bands resulting from κ-carbide precipitation.[40] Interestingly, as the amount of age hardening was increased to 30 h, the difference in the −40°C breaking energy was statistically the same regardless of silicon content.

The as-cast solution-treated and aged CVN notch toughness was compared with the room-temperature notch toughness of a hot-rolled Fe-30Mn-9Al-2.7Si-0.9C steel with 0.023% P and an average grain size of 50 μm. In the as-hot-rolled condition, the breaking energy was only 23 J in the L–T orientation and 22 J in the T–L orientation. The fracture surface of the as-hot-rolled steel is shown in Fig. 13a to consist mainly of transgranular cleavage-type fracture with little evidence of ductility.

The cleavage facets displayed slip traces at roughly 60° angles from one another indicating fracture by decohesion of active slip planes. Solution treating for 10 min at 1100°C and water quenching greatly improved toughness to an average of 160 J (L–T orientation), and fracture was completely ductile in nature as shown in Fig. 13b. This large difference in energy between hot-rolled

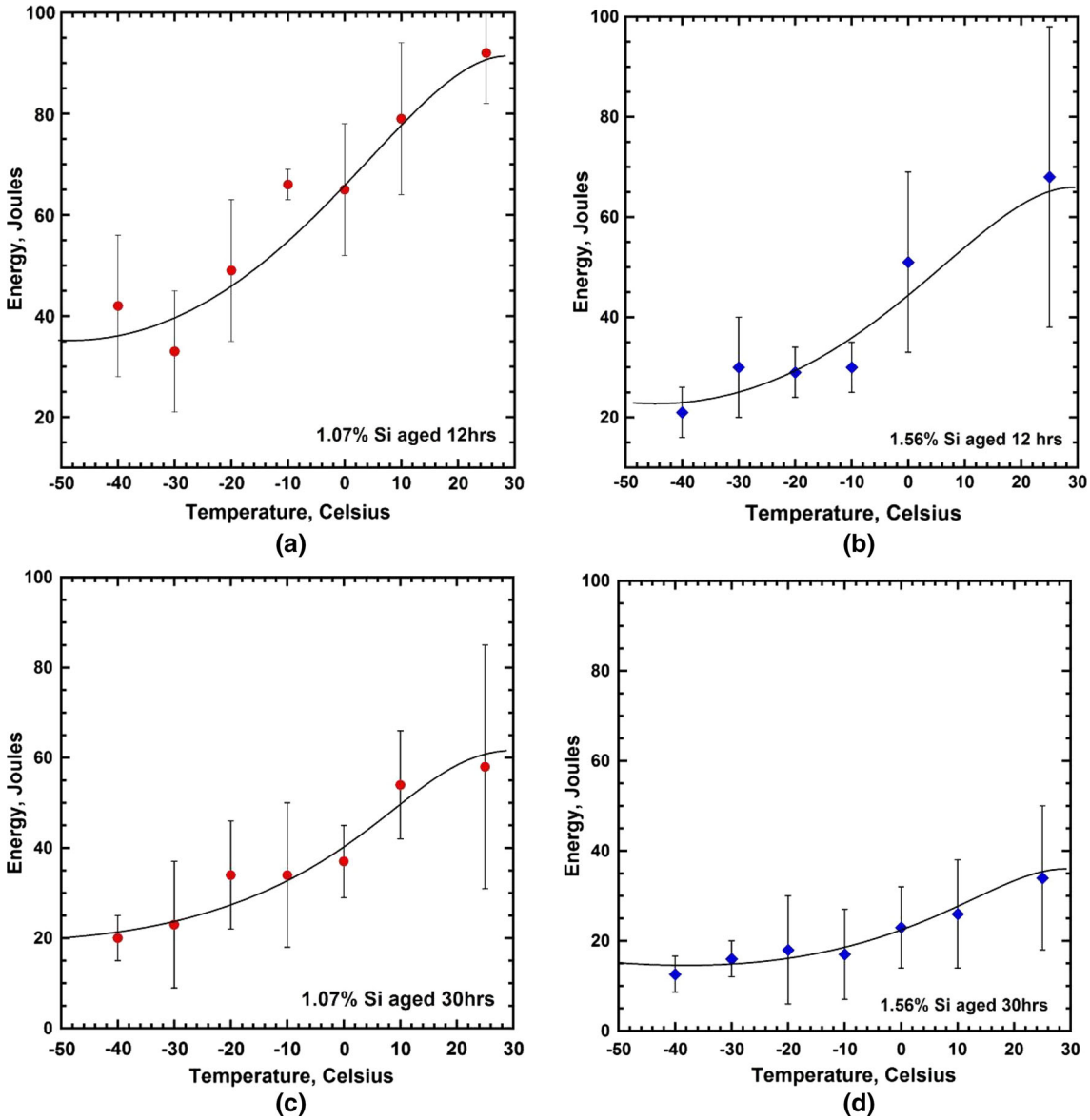


Fig. 11. Charpy V-notch impact energy as a function of testing temperature for Fe-30Mn-9Al-0.9C alloy specimens containing 1.07 wt.% Si aged from 12 h to 30 h at 530°C (a and c) and curves for specimens containing 1.56 wt.% Si aged from 12 h to 30 h at 530°C (b and d). As the amount of silicon increased, the breaking energy was found to decrease for all temperatures and aging conditions. Increased κ-carbide precipitation was shown to dramatically decrease the impact energy for both silicon level samples.

and the hot-rolled and solution-treated condition is related to SRO and precipitation of κ-carbide because the hot-rolled material was slow cooled after deformation.

DYNAMIC FRACTURE TOUGHNESS

Notch toughness of high-strength steel has traditionally been characterized by the CVN breaking energy, which is only a qualitative measure of toughness under impact loading and is geometry specific. To qualify lightweight Fe-Mn-Al-C steels for high energy-absorbing applications, knowledge of the high-strain-rate fracture toughness, or dynamic fracture toughness, in the presence of a sharp crack is desired. For ductile materials exhibiting crack tip blunting before stable crack propagation, an elastic plastic fracture mechanics (EPFM) approach is necessary and the flow characteristics can be described by the path-independent J integral, which is equivalent to the energy release rate in elastic–plastic materials.[41] The dynamic fracture toughness (DFT) of ductile materials is therefore defined as the critical value of the J integral near the onset of stable crack growth.[42] Instrumented Charpy impact tests on fatigue precracked specimens provide a reproducible way of measuring the time dependency of force and crack displacement at elevated loading rates, thus providing a means of measuring DFT. Schindler[43] has

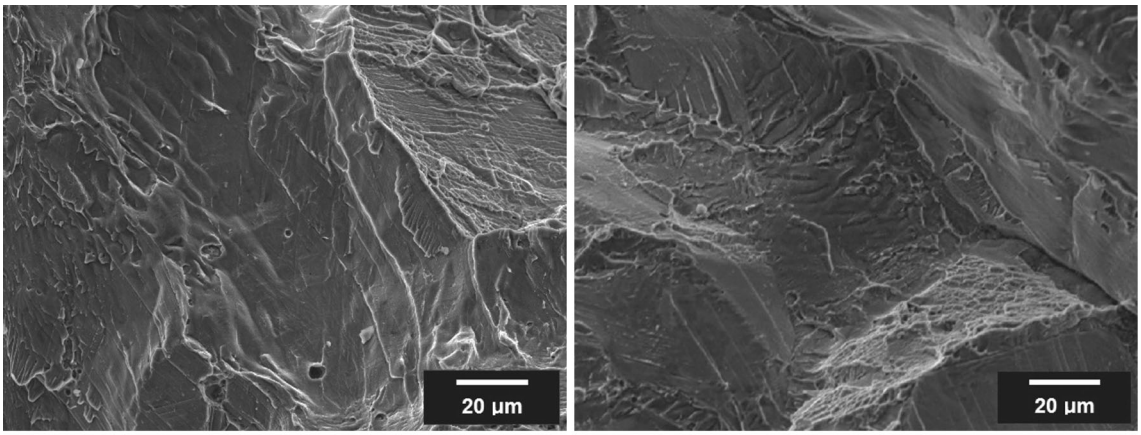


Fig. 12. The fracture surfaces of the 12 h aged (a) 1.07% Si and (b) 1.56% Si specimens that were broken at −40°C show mostly transgranular cleavage-like fracture with slip traces noted on certain favorably oriented cleavage facets.

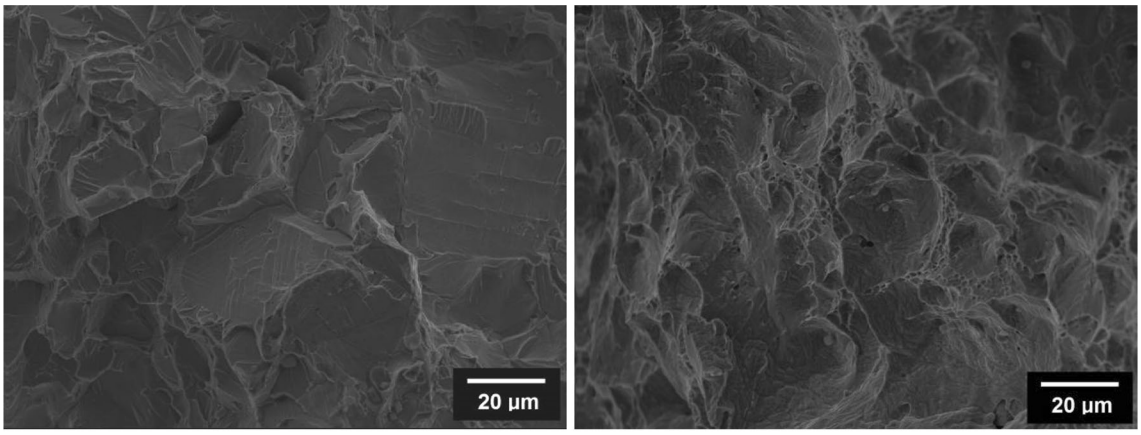


Fig. 13. The CVN fracture surfaces of the (a) as-hot-rolled and (b) hot-rolled, solution-treated, and quenched Fe-30Mn-9Al-2.7Si-0.9C steel specimens. (a) In the as-hot rolled specimens, fracture was mainly transgranular cleavage with {111} slip traces on cleavage facets indicating brittle fracture due to κ-carbide precipitation. (b) After solution treatment for 10 min at 1100°C, the fracture mode changes to a completely ductile mode.

devised a single-specimen technique for determining the dynamic fracture toughness J_{Id} from instrumented Charpy tests. The only experimental inputs necessary for this technique are the peak load (P_{max}), the energy consumed up to the peak load (E_{max}), and the total fracture energy (E_{tot}), all of which can be determined in a straightforward fashion from the instrumented load versus crack displacement data. A complete description of the test procedure and analysis of the data may be obtained from Schindler,[43] ASTM E 1820,[42] and ASTM E 813.[44] The Schindler[43] method was used to determine the effects of silicon, phosphorus, and aged hardness on the dynamic fracture toughness of a Fe-30Mn-9Al-XSi-0.9C alloy. The results are shown in Table IV and are correlated with DFT values for quench and tempered cast 4130 steel.

In the solution-treated condition, the dynamic fracture toughness was statistically equivalent between the low phosphorus (<0.003% P) 1.07 and 1.56% Si specimens with an average value around 700 kJ/m^2. Age hardening sharply reduced the toughness and transitioned the fracture mode from ductile failure to brittle cleavage-like fracture characterized by {111} slip traces on favorably oriented cleavage planes as shown in Fig. 14. Increasing the amount of silicon from 1.07% to 1.56% Si decreased the DFT of age-hardened steels and increased the propensity for brittle fracture.

Brittle fracture is thus attributed to an increase in the size, carbon content, and perhaps volume fraction of κ-carbide during aging with increasing amounts of silicon. The formation of planar slip bands in age-hardenable Fe-Mn-Al-C steels is related to the shearing of κ-carbides with a corresponding loss in strain hardening and toughness. Increases in the size, volume fraction, or carbon content of κ-carbide during aging increases the

Table IV. Dynamic fracture toughness of Fe-30Mn-9Al-XSi-0.9C steels

Alloy	Heat treatment	Hardness (BHN)	DFT (kJ/m^2)
0.59% Si, 0.002% P	Solution Treated at 1050°C 2 h	200 ± 10	715
	Aged at 530°C 22 h	304 ± 13	242 ± 82
	Aged at 530°C 90 h	366 ± 9	95 ± 15
1.07% Si, 0.001% P	Aged at 530°C 13 h	295 ± 14	376 ± 23
	Aged at 530°C 60 h	358 ± 8	153 ± 32
1.56% Si, 0.002% P	Solution Treated at 1050°C 2 h	208 ± 10	700
	Aged at 530°C 10 h	318 ± 14	265 ± 66
	Aged at 530°C 40 h	375 ± 6	144 ± 50
0.94% Si, 0.043% P	Aged at 530°C 10 h	340 ± 10	100
Cast 4130	Quenched and tempered	350	94
	Quenched and tempered	330	136

Uncertainty reported for test sample greater than four and represents one sample standard deviation

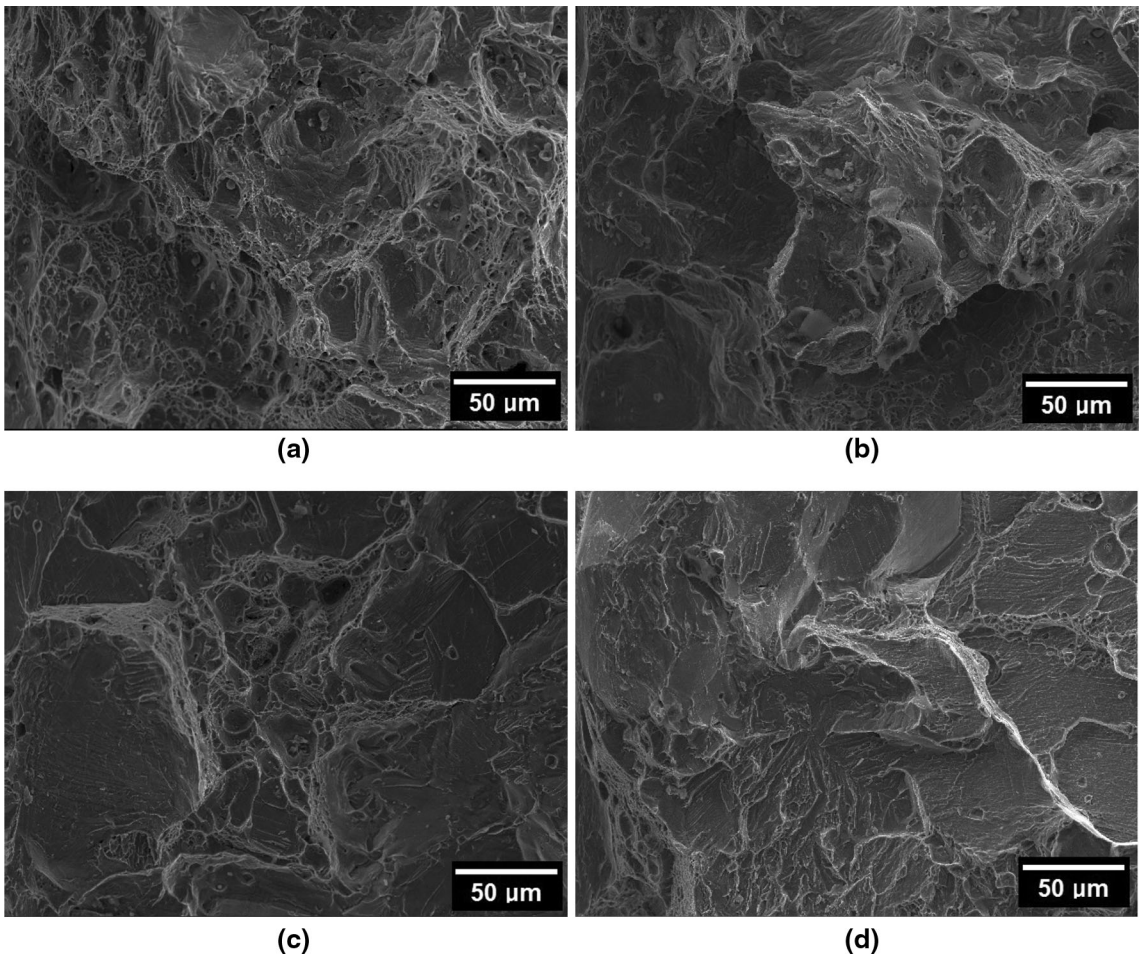


Fig. 14. The fracture surfaces of the Fe-30Mn-9Al-0.9C DFT specimens with (a and c) 1.07% Si and (b and d) 1.56% Si aged at 530°C to a hardness in the range of 300 BHN (a and b) and 350 BHN (c and d) show a transition from ductile failure to brittle transgranular cleavage-type fracture with increased age hardening and silicon content. (c and d) {111} slip traces are noted on favorably oriented cleavage facets suggesting fracture by decohesion of glide planes.

likelihood of planar slip being concentrated on relatively few slip bands as the carbides are sheared during plastic deformation. As silicon has been shown to increase the size and carbon content of κ-carbide during aging, this would also contribute to an increase in dislocation accumulation on

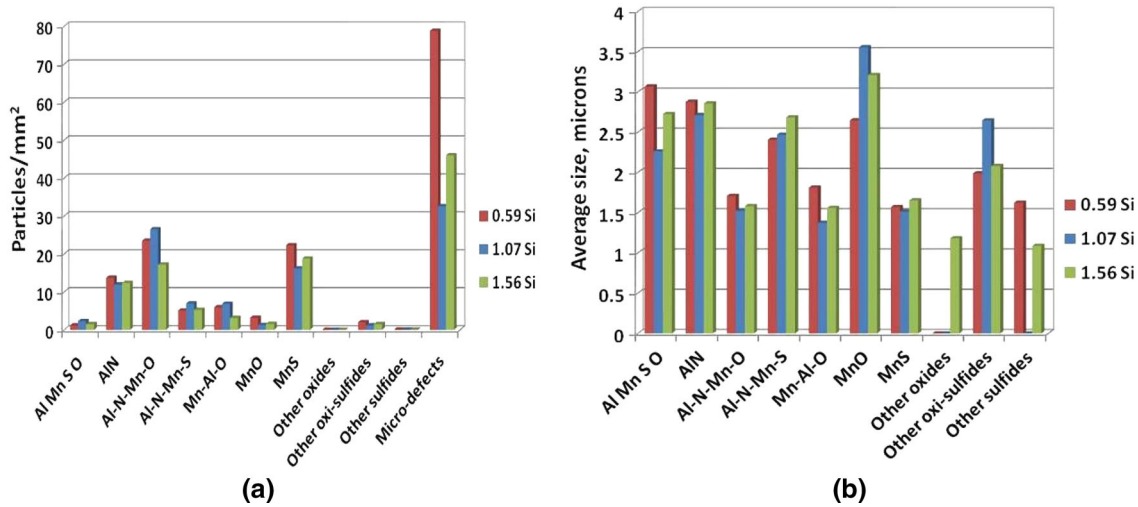


Fig. 15. (a) The average inclusion density by type of inclusion shows that the number of inclusions is very similar between the silicon-containing steels, and the only difference is the much larger amount of porosity in the 0.59% Si steel when compared with the others. (b) The average inclusion size was found to be largely similar between the different steels.

relatively few slip bands and increases the propensity of {111} decohesion, especially at high strain rates. Curiously, the age-hardened 0.59% Si specimens (0.002% P) were observed to have the lowest toughness values of the low-phosphorus alloys as shown in Table IV. As previously mentioned, for a given heat treatment, toughness is a function of chemistry and inclusion content, especially AlN. Inclusion analysis was performed using an automated feature analysis, and these results are shown in Fig. 15. The total inclusion density was similar between the three low-phosphorus silicon-containing steels with between 62 and 78 inclusions/mm^2. Inspection of Fig. 15 shows that the only major difference in inclusion content between the different silicon containing steels was the amount of porosity. The porosity in the 0.59% Si steel was more than twice that of the 1.07% Si steel and almost double that of the 1.56% Si steel. In addition to AlN, porosity in high-strength steel has been shown to deleteriously affect notch toughness and was reported to decrease the toughness of quenched and tempered Cr and Mo steel by more than 50 kJ/m^2.[45] It is therefore likely that the lowered toughness of the aged 0.59% Si sample is the result of an increased amount of porosity.

The dynamic fracture toughness of an aged high-phosphorus specimen, 0.043% P, with 0.94% Si was reported to be 100 kJ/m^2 and was much lower than the 1.07% Si specimens aged for the same approximate times at 530°C and fracture produced a dendritic pattern with cleavage along the {100}. This has been attributed to preferential and kinetically faster phosphorus substitution for aluminum positions in the κ-carbide with a predicted 45% reduction in the {100} cleavage stress.[29] The nature of the brittle fracture in these lightweight steel is thus dependent on phosphorus content, SRO, and planar slip induced by the shearing of κ-carbide.

The toughness of these alloys would be greatly enhanced by a reduction in grain size. Additional research is required to develop grain-refining agents for these cast alloys. Grain refinement by forging is a possibility where cast preforms are reduced and recrystallized during solution treatment. Furthermore, there is a need to develop dispersoids capable of interfering with the planar slip induced by either SRO or shearing of κ-carbide to improve notch toughness and fatigue properties.

CONCLUSION

High manganese and aluminum advanced high-strength steels (AHSS) continue to be of interest to the military, mining, and transportation industries because of an excellent combination of high-strain-rate toughness coupled with up to a 15% reduction in density. The tensile and high-strain-rate fracture properties associated with austenitic Fe-30%Mn-9%Al-0.9%C steels of varying silicon contents (0.59–1.56% Si) and impurity levels of phosphorus (0.001–0.07% P) were reviewed. In the solution-treated condition and for steels with low phosphorus (< 0.007% P), these steels have high work-hardening rates with strengths up to 737 MPa and dynamic fracture toughness over 700 kJ/m^2. Silicon additions in the range of 0.59–1.56% Si were found to have no significant effect on the mechanical properties of solution-treated steels but increased the tensile strength and hardness during aging at 530°C. Dynamic fracture toughness in age-hardened steels decreased with silicon addition. Silicon has been shown to increase the kinetics of κ-carbide precipitation by increasing the activity of carbon in austenite and thus increasing the partitioning of carbon into the κ-carbide during aging. As the addition of silicon was increased from 1.07% to 1.56% in age-hardened steels, the Charpy V-notch

toughness decreased from 92 J to 68 J and the dynamic fracture toughness decreased from 376 kJ/m^2 to 265 kJ/m^2, respectively. Toughness was found to be a strong function of phosphorus, decreasing the solution treated CVN impact toughness from 200 J in a 0.006% P steel to 28 J in a 0.07% P steel with a nominal silicon content of 1% Si. For aged steels with nominally 1% Si, increasing the levels of phosphorus from 0.001% to 0.043% decreased the dynamic fracture toughness from 376 kJ/m^2 to 100 kJ/m^2. It is shown from this review that lightweight steels possess many attributes that may facilitate the light weighting of transportation vehicles. These steels can be processed to obtain mechanical properties equivalent to Cr-Mo steels but at a weight savings of 15%. These lightweight steels also exhibit higher dynamic fracture toughness, which may be useful in improving crash worthiness and thus the safety of vehicle passengers. Further improvement to these steels will require grain refining techniques to reduce the cast austenite grain size or cast preforms that can subsequently be grain refined by forging. It is doubtful that these high-manganese (>25% Mn) steels will be produced by standard steelmaking techniques and production will be restricted to the foundry industry. In that regard, sufficient research has been accomplished to provide guidance in the production of the lightweight steels.

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